

A Comparative Benchmark of Semantic, Statistical, and Structural Retrieval Paradigms for Retrieval-Augmented Generation over SEC 10-K Filings

Design and Specification Proposal, COMP702 M.Sc. Project (2026/27)

Submitted by

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Statement of Ethical Compliance: A0

Data Category: A. Every source document is a United States SEC 10-K filing obtained from SEC EDGAR, a public regulatory archive. The data is publicly available, contains no personal or confidential information, and its use requires no licence, consent, or data-sharing agreement.

Participant Category: 0. The project involves no human participants at any stage. The 140-query benchmark is generated and verified entirely by two independent language models acting as generator and critic, and the two manual steps (labelling teaching exemplars and hand-scoring validation outputs) are carried out solely by the researcher on the project's own outputs, not on data gathered from, or about, other people.

I confirm that I have read the ethical guidelines and will follow them during this project. This project uses only publicly available, non-personal United States SEC 10-K corporate filings sourced from SEC EDGAR. It involves no human participants and no personal data of any kind.

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1. Project Description

Retrieval-Augmented Generation (RAG) systems answer questions by pairing a retriever that selects relevant evidence from a corpus with a language model that generates a grounded answer (Lewis *et al.*, 2020). Most RAG benchmarks target encyclopaedic or open-domain text; far less is known about how different retrieval paradigms behave on dense, table-heavy financial disclosures such as SEC 10-K filings, where an answer may sit inside a single cell of a multi-row table rather than a sentence of prose (Chen *et al.*, 2021). This project builds and benchmarks three complete RAG pipelines: a semantic retriever, which finds evidence by meaning using numeric representations of text (dense vector embeddings); a statistical retriever, which finds evidence by matching exact words and their frequency using the long-established BM25 algorithm; and a structural retriever, which navigates a tree of AI-generated summaries built once over each document. These three are tested against a 140-query, human-anchored benchmark spanning direct and inferential questions over both narrative text and financial tables, to establish where each paradigm succeeds and where it degrades.

2. Aims and Requirements

2.1 Aims

- Design and implement three architecturally distinct RAG retrieval pipelines (semantic, statistical, structural) sharing one evaluation harness, so that performance differences are attributable to retrieval strategy rather than to generation.
- Construct an adversarially verified, human-anchored benchmark of 140 questions, stratified across four query types (direct/implicit retrieval over text/table content), grounded in real SEC 10-K filings.
- Quantify each pipeline's retrieval accuracy, answer quality, and computational efficiency, and identify the conditions under which each paradigm degrades.

2.2 Requirements: Essential

- Parse SEC 10-K filings into metadata-enriched, atomic text/table nodes without bisecting tables.
- Implement three distinct pipelines - P1 (dense embeddings with a cross-encoder reranker), P2 (Okapi BM25), and P3 (hierarchical summary-tree) retrievers sharing one answering model.

- Generate and adversarially verify a 140-query benchmark with node level ground truth citations.
- Validate an automated LLM judge against human scores before using it to grade the full benchmark.
- Persist all queries, retrieved evidence, and scores in a relational (SQLite) store with crash-safe resume.

2.3 Requirements: Desirable

- Extend the comparison with a metadata pre-filter or hybrid-fusion ablation (reserved future-scope pipelines).
- Visualise per-quadrant performance gaps in a comparison view for the dissertation write-up.

3. Key Literature and Background Reading

Lewis *et al.* (2020) framed answer generation grounded in retrieved evidence as a way of reducing hallucination relative to purely parametric generation, and this Retrieval-Augmented Generation framing motivates the project's overall design: a retriever supplies evidence, and one shared answerer consumes it.

On the statistical side, Robertson and Zaragoza (2009) formalised the probabilistic relevance framework that underlies BM25, which remains a strong, fully deterministic retrieval baseline and forms this project's P2 pipeline. On the semantic side, Karpukhin *et al.* (2020) showed that learned dense bi-encoder retrieval can outperform BM25 by 9–19% absolute on open-domain question answering, while Nogueira and Cho (2019) showed that adding a cross-encoder reranking stage over an initial candidate set materially improves precision; together these findings motivate the project's P1 embed-then-rerank design. The embedding and reranker models used for P1 draw on the BGE family released by Xiao *et al.* (2023).

Sarathi *et al.* (2024) introduced RAPTOR, which recursively clusters and summarises text into a retrievable tree; this is the architectural basis for the project's P3 structural pipeline, although their evaluation, like most RAG literature, is conducted on narrative rather than tabular text.

The domain challenge that motivates this project is well documented: Chen *et al.* (2021) introduced FinQA and showed that even strong pre-trained models fall far short of expert humans at multi-step numerical reasoning over financial tables, which is why the benchmark constructed here

explicitly stratifies table-heavy query quadrants. Liu *et al.* (2024) further showed that model accuracy degrades when relevant evidence sits in the middle of a long context, which is the argument for retrieval-based context selection over naive full-document stuffing.

Finally, on evaluation methodology, Zheng *et al.* (2023) showed that a strong LLM judge can reach over 80% agreement with human preference judgements, the same threshold this project adopts as its judge-validation gate before trusting an automated judge to grade the full benchmark.

None of the studies above evaluate all three retrieval paradigms side by side on a single financial-filing benchmark stratified by both query type and content modality (text versus table); this is the specific gap the proposed project addresses.

4. Development and Implementation Summary

The system is implemented in Python 3, using LlamaIndex as a common retrieval and orchestration backbone so that all three pipelines expose the same retriever interface. Python is the natural choice because the entire required ecosystem, LlamaIndex, the Groq SDK, rank_bm25, ChromaDB, and the HuggingFace BGE models, is Python-native, so reimplementing any part of it in another language would mean rebuilding mature retrieval and embedding infrastructure from scratch. LlamaIndex specifically is chosen over a lower-level alternative such as LangChain because its node and scored-result abstractions already match the project's own node-centric data model (Section 5), allowing the three pipelines to share one retriever interface without an additional adapter layer. Generation, dataset critique, and judging are performed by open-source instruction models served free of charge via Groq (rate-limited rather than billed, so the implementation has zero recurring spend). Embeddings (BAAI/bge-small-en-v1.5) and reranking (BAAI/bge-reranker-base) run locally on CPU.

ChromaDB provides the local vector store for P1, rank_bm25 provides the keyword index for P2, and a cached LlamaIndex SummaryIndex provides the structural pipeline for P3. A single SQLite database, run in Write-Ahead Logging mode, is the sole state store; every benchmark run commits immediately, so a crash resumes from the last written row rather than re-spending free-tier quota. Queries are split into three disjoint sets, PQ (100, the benchmark), GQ (20, Judge teaching exemplars) and JEQ (20, Judge validation), enforced at the schema level so no pipeline or prompt ever sees beyond its own group. Asynchronous worker pools, bounded by a semaphore and wrapped in an exponential-backoff retry layer, keep concurrent API calls within Groq's free-tier rate limits. Workflow is organised phase-by-phase as set out in Section 10, with the code for each

phase committed and exercised under a three-item local throttle before being released at full scale, so a fault discovered in one phase is never allowed to consume the free-tier quota reserved for an earlier or later phase.

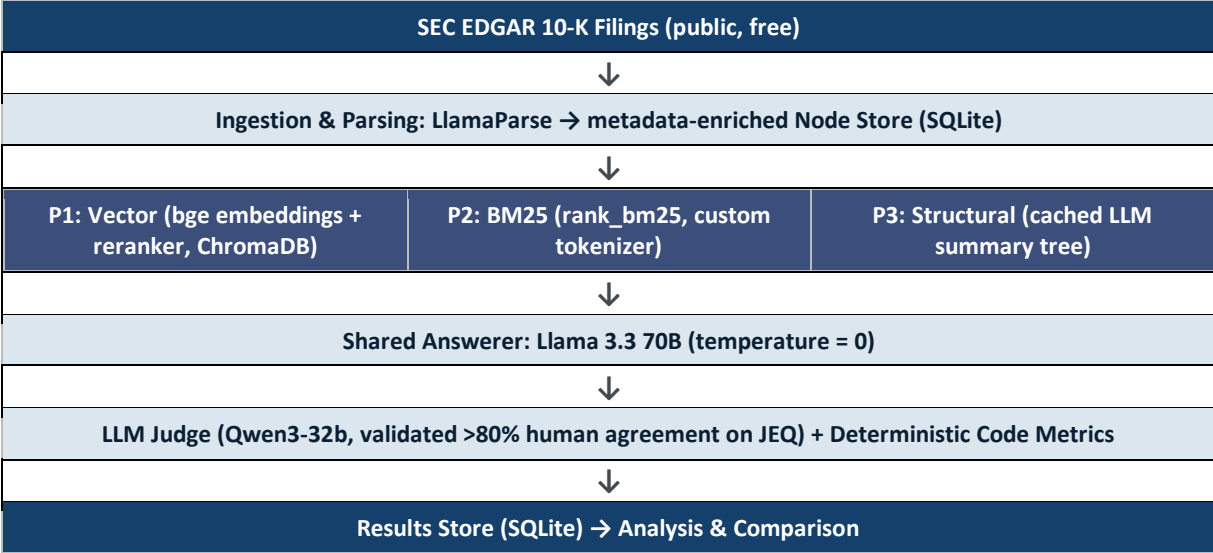


Figure 4.1: High-Level Pipeline Architecture

Key point. Every recurring component in this pipeline runs on a free tier or local CPU, so the expected spend for the full benchmark is \$0.00; the only binding constraint is Groq's daily token throughput, not budget.

5. Data Sources

All data is sourced from SEC EDGAR, the US Securities and Exchange Commission's public filing system: a corpus of approximately six to nine 10-K annual filings, drawn from two to three companies across two to three fiscal years. Filings are parsed via LlamaParse (free tier) into metadata-enriched text and table nodes.

5.1 Nature of the SEC 10-K Filing Data

Each filing contains narrative sections (risk factors, business description, management's discussion) and structured tables (financial statements, segment data, numerical metrics with currency values). The data is heterogeneous: prose sections use formal language with embedded financial figures, while tables present precisely formatted multi-year numerical comparisons. This heterogeneity is valuable for benchmarking. It tests whether retrieval pipelines can handle both semantic and structured data extraction. The raw structure of the data is covered in the next section (5.2).

5.2 Raw Data Sample of the SEC 10-K Filing Data

Figure 5.1 shows a sample paragraph from Apple's SEC 10-K filing in raw iXBRL format, with a financial figure embedded in XML markup. Figure 5.2 displays the rendered table corresponding to the raw HTML/XML code in Figure 5.3. This table from Apple's 10-K, presents interest rate sensitivity analysis with comparative figures across fiscal years. Together, these samples illustrate the dual nature of SEC 10-K data: unstructured prose with embedded financial figures and structured tabular data with precise column and row definitions. Both formats are parsed as nodes during data preparation, preserving their original structure and metadata for retrieval and evaluation.

```
<div style="margin-top:9pt;text-align:justify">
  <span style="color:#000000;font-family:'Helvetica',sans-serif;font-size:8pt;font-
weight:400;line-height:120%">
    The aggregate market value of the voting and non-voting stock held by
    non-affiliates of the Registrant, as of March&#160;28, 2025, the last business
    day of the Registrant&#8217;s most recently completed second fiscal quarter,
    was approximately $
    <ix:nonFraction unitRef="usd" contextRef="c-10" decimals="-6"
name="dei:EntityPublicFloat"
    format="ixt:num-dot-decimal" scale="0" id="f-50">
      3,253,431,000,000
    </ix:nonFraction>
    . Solely for purposes of this disclosure, shares of common stock held
    by executive officers and directors of the Registrant as of such date have
    been excluded because such persons may be deemed to be affiliates. This
    determination of executive officers and directors as affiliates is not
    necessarily a conclusive determination for any other purposes.
  </span>
</div>
```

Figure 5.1: Sample paragraph data structure (iXBRL disclosure section)

Interest Rate Sensitive Instrument	Hypothetical Interest Rate Increase	Potential Impact	2025	2024
Investment portfolio	100 basis points, all tenors	Decline in fair value	\$2,416	\$2,755
Term debt	100 basis points, all tenors	Increase in annual interest expense	\$ 129	\$ 139

Image 5.2: Sample table render (Interest Rate Sensitivity table)


```

<table style="border-collapse:collapse;width:100%">
  <tr>
    <td colspan="3" style="padding:2px 1pt;font-weight:700">
      <div>Interest Rate</div>
      <div>Sensitive Instrument</div>
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt;font-weight:700">
      <div>Hypothetical Interest</div>
      <div>Rate Increase</div>
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt;font-weight:700">
      <div>Potential Impact</div>
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt;text-align:center;font-weight:700">2025</td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt;text-align:center;font-weight:700">2024</td>
  </tr>
  <tr>
    <td colspan="3" style="border-top:1pt solid #000;padding:2px 1pt">
      Investment portfolio
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="border-top:1pt solid #000;padding:2px 1pt">
      100 basis points, all tenors
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="border-top:1pt solid #000;padding:2px 1pt">
      <div>Decline in fair value</div>
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td style="border-top:1pt solid #000;padding:2px 0 2px 1pt">$</td>
    <td style="border-top:1pt solid #000;padding:2px 0;text-align:right">2,416</td>
    <td style="border-top:1pt solid #000;padding:2px 1pt 2px 0" />
    <td colspan="3" style="padding:0 1pt" />
    <td style="border-top:1pt solid #000;padding:2px 0 2px 1pt">$</td>
    <td style="border-top:1pt solid #000;padding:2px 0;text-align:right">2,755</td>
    <td style="border-top:1pt solid #000;padding:2px 1pt 2px 0" />
  </tr>
  <tr style="background-color:#efefef">
    <td colspan="3" style="padding:2px 1pt;padding-left:11pt">
      Term debt
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt 2px 10pt">
      100 basis points, all tenors
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td colspan="3" style="padding:2px 1pt 2px 10pt">
      Increase in annual interest expense
    </td>
    <td colspan="3" style="padding:0 1pt" />
    <td style="padding:2px 0 2px 1pt">$</td>
    <td style="padding:2px 0;text-align:right">129</td>
    <td style="padding:2px 1pt 2px 0" />
    <td colspan="3" style="padding:0 1pt" />
    <td style="padding:2px 0 2px 1pt">$</td>
    <td style="padding:2px 0;text-align:right">139</td>
    <td style="padding:2px 1pt 2px 0" />
  </tr>
</table>

```

Figure 5.3: Sample table data structure (Interest Rate Sensitivity table)

6. Testing and Evaluation

The system is evaluated across three independent pillars:

- **Retrieval metrics** (Precision@K, Recall@K, Evidence Hit Rate) are computed deterministically against node-ID ground truth.
- **Answer-quality metrics** use an LLM-as-judge score (1–10) as the primary measure, with token-level F1 and numeric-tolerant Exact Match as secondary lexical checks, plus a code-level Citation Audit that flags answers which are correct only by coincidence, where the cited evidence does not match the ground truth.
- **Efficiency metrics** cover latency, token consumption, and one-time index-build cost per pipeline.

The automated judge is not trusted by default. It must first clear a hard validation gate, reaching greater than 80% agreement with the researcher’s own hand-scored outputs on a disjoint validation set, before it is permitted to grade the full benchmark, mirroring the agreement threshold reported for LLM judges by Zheng *et al.* (2023). Every script that calls a language model is first run end-to-end under a hardcoded three-item local throttle, to catch bugs before any free-tier quota is spent at scale.

Key point. The judge is gated, not trusted by default: it must clear 80% agreement with hand-scored human judgements on a disjoint validation set before it is allowed to grade the full 900-run benchmark.

7. Project Ethics and Human Participants

This project involves no human participants and no personal data; all source material is SEC 10-K filings, which are public corporate disclosures (Data Category A0). The two manual steps in the methodology, hand-labelling 20 teaching exemplars and hand-scoring 60 judge-validation outputs, are carried out solely by the researcher as part of constructing the evaluation harness itself, rather than as a study involving external participants, and therefore fall outside the scope of human-participant ethics review. No further ethical approval beyond this statement is required for the activity planned within this assessment.

8. BCS Project Criteria

- **Systematic understanding:** a cross-paradigm RAG comparison grounded in current literature (Lewis et al., 2020; Karpukhin et al., 2020; Sarthi et al., 2024), exposing paradigm limits not yet documented for SEC-filing data.
- **Comprehensive technique understanding:** dense retrieval, sparse retrieval, hierarchical summarisation, reranking, and LLM-based evaluation are all designed and built within a single project.
- **Originality:** the cross-family generator/critic adversarial verification pipeline and the disjoint three-set (PQ/GQ/JEQ) evaluation design are original contributions to constructing a RAG benchmark without manual annotation at scale.
- **Sound judgement under incomplete information:** deliberate, documented scoping decisions, such as dropping a metadata pre-filter and accepting a token-per-day-bound runtime, demonstrate judgement under real resource constraints.
- **Self-direction:** a one person, ten weeks build plan run entirely on free-tier infrastructure with no managed budget demonstrates autonomous planning.
- **Critical self-evaluation of the process:** the Limitations framing already built into the methodology, a single temperature-zero run per benchmark cell and an approximately eighty per cent judge-validation pass on only sixty hand-scored samples, is treated explicitly as a pragmatic constraint rather than a strong statistical proof.

9. UI/UX Mockup

This is a research and benchmarking system rather than an end-user application, so its principal "interface" is the results comparison view produced at the end of the run. Day to day, the researcher's surface is a command line. A wireframe sketch of the final comparison view, the artefact consulted most, is shown below.

Pipeline	K	Precision@K	Recall@K	Judge Score	Latency (s)
P1_vector	5	0.62	0.81	8.1	1.84
P2_bm25	5	0.54	0.73	7.0	0.91
P3_structural	5	0.41	0.66	6.4	1.10

Figure 9.1: Results Comparison View (Wireframe, illustrative placeholder values, not final results)

10. Project Plan

The build is scoped to a one person, ten-week schedule, with active development substantially complete by Week 7 and the final three weeks overlapping the full benchmark run with results analysis, dissertation write-up, and preparation of the final project presentation.

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1. Environment & Infra										
2. Ingestion & Parsing										
3. P3 Summary-Index Build										
4. Dataset Gen & Verification										
5. Pipeline Impl. (P1/P2/P3)										
6. Judge Build & Validation Gate										
7. Full Benchmark Execution										
8. Results Analysis & Write-up										

Table 10.1: 10-Week Project Gantt Chart

11. Risks and Contingency Plans

Risk	Contingency	Likelihood	Impact
Groq free-tier token-per-day limit reached mid-run	Exponential backoff with jitter, multi-day paced execution with crash-safe resume, prompt caching, fallback to the paid Developer tier if the schedule slips	Medium	Medium
LlamaParse fragments or bisects a financial table	Randomised 20-section manual parsing audit; re-tune parser settings or discard and re-parse the affected filing	Medium	High
Automated LLM judge fails the 80% human-agreement gate	Tune the scoring rubric and/or swap the judge to a different model family; re-run the validation gate before proceeding	Medium	High
Breaking API changes in a fast-moving dependency (LlamaIndex)	Pin exact package versions at project start; consult current documentation before any upgrade	Low	Medium
Concurrent writes corrupt the SQLite results store mid-run	Write-Ahead Logging (WAL) mode; crash-safe resume keyed on a unique constraint; commit after every individual run	Low	High
One person, ten-week timeline slips	Throttle-first testing on every phase before full-scale release; background-paced benchmark execution; buffer reserved across the final three weeks	Medium	Medium

Risk	Contingency	Likelihood	Impact
Local machine failure or loss of unsynced work (no backups)	Nightly off-machine backup of the SQLite database, parsed nodes, and code to a second location (a git remote and cloud sync); Write-Ahead Logging commits keep the loss window to minutes of work, not days	Low	High

Table 11.1: Risk Management Plan

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